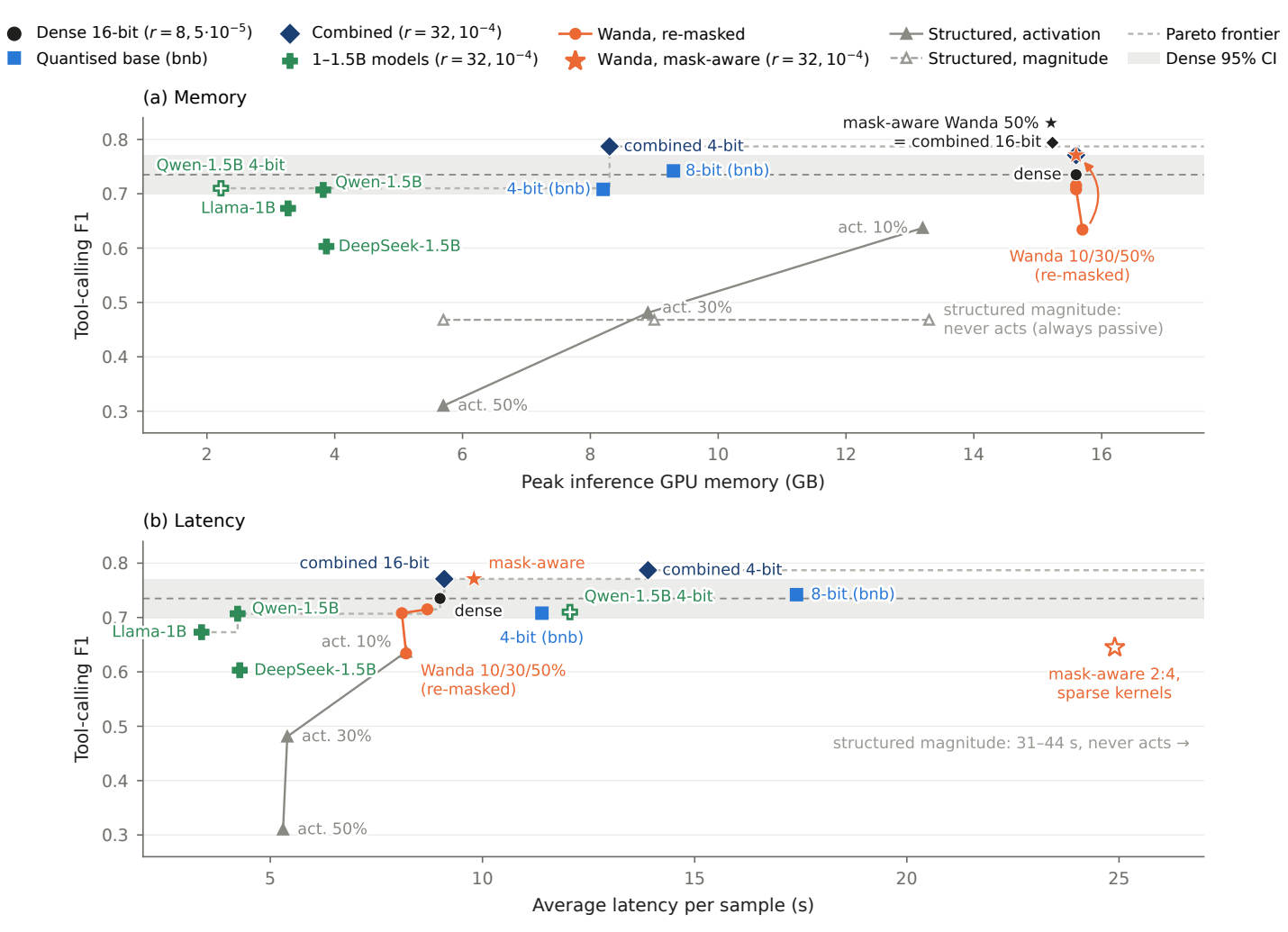

Dense 16-bit ($r = 8, 5{\cdot}10^{-5}$)
Quantised base (bnb)
Combined ($r = 32, 10^{-4}$)
1–1.5B models ($r = 32, 10^{-4}$)
Wanda, re-masked
Wanda, mask-aware ($r = 32, 10^{-4}$)
Structured, activation
Structured, magnitude
Pareto frontier
Dense 95% CI
(a) Memory
Tool-calling F1
Peak inference GPU memory (GB)
mask-aware Wanda 50% ★
= combined 16-bit ◆
combined 4-bit
8-bit (bnb)
dense
Qwen-1.5B 4-bit
Qwen-1.5B
4-bit (bnb)
Llama-1B
DeepSeek-1.5B
act. 10%
Wanda 10/30/50%
(re-masked)
act. 30%
structured magnitude:
never acts (always passive)
act. 50%
(b) Latency
Average latency per sample (s)
combined 16-bit
mask-aware
combined 4-bit
Qwen-1.5B 4-bit
8-bit (bnb)
Qwen-1.5B
dense
Llama-1B
4-bit (bnb)
act. 10%
DeepSeek-1.5B
Wanda 10/30/50%
(re-masked)
mask-aware 2:4,
sparse kernels
act. 30%
structured magnitude: 31–44 s, never acts →
act. 50%